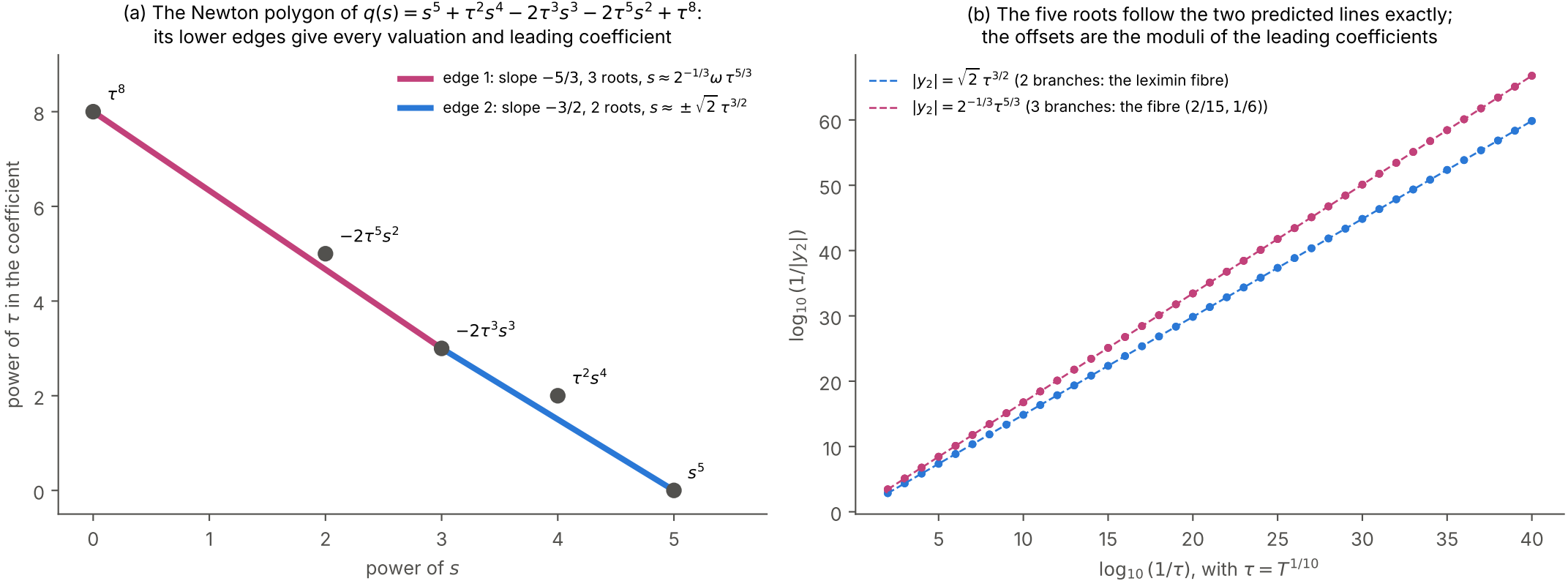
(a) The Newton polygon of $q(s) = s^5 + \tau^2 s^4 - 2\tau^3 s^3 - 2\tau^5 s^2 + \tau^8$:
its lower edges give every valuation and leading coefficient
edge 1: slope −5/3, 3 roots, $s \approx 2^{-1/3}\omega\,\tau^{5/3}$
edge 2: slope −3/2, 2 roots, $s \approx \pm\sqrt{2}\,\tau^{3/2}$
τ^8
$-2\tau^5 s^2$
$-2\tau^3 s^3$
$\tau^2 s^4$
s^5
power of τ in the coefficient
power of s
0
2
4
6
8
0
1
2
3
4
5
(b) The five roots follow the two predicted lines exactly;
the offsets are the moduli of the leading coefficients
$|y_2| = \sqrt{2}\,\tau^{3/2}$ (2 branches: the leximin fibre)
$|y_2| = 2^{-1/3}\tau^{5/3}$ (3 branches: the fibre (2/15, 1/6))
$\log_{10}(1/|y_2|)$
$\log_{10}(1/\tau)$, with $\tau = T^{1/10}$
0
10
20
30
40
50
60
5
10
15
20
25
30
35
40